

Academic Programs

PEOs, POs & PSOs

B.Tech ECE

Program Educational Objectives (PEOs)

The student in Electronics and Communication Engineering

will be able to:

Produce Electronics and Communication Engineering

Professionals with a solid foundation in Mathematics,
Science and Technology which is essential to solve
engineering problems.

Train students in good scientific and engineering practices

so that they comprehend, analyze, design, and create novel
products and offer solutions for industry specific processes
and real life problems.

Prepare students to adopt the learning culture needed for a

successful professional career by encouraging them to
acquire higher qualifications, take up research and keep
abreast of latest technological developments.

Inculcate organizing, managerial and entrepreneurship skills

essential for professional growth.

Develop the consciousness among students towards moral

values and professional ethics while developing innovative
solutions to meet the societal needs.

Program Outcomes (POs)

Engineering Graduates will be able to:

Engineering knowledge: Apply the knowledge of mathematics,
science, engineering fundamentals, and an engineering
specialization to the solution of complex engineering
problems.

Problem analysis: Identify, formulate, review research

literature, and analyze complex engineering problems
reaching substantiated conclusions using first principles of
mathematics, natural sciences, and engineering sciences.

Design/development of solutions: Design solutions for

complex engineering problems and design system components or
processes that meet the specified needs with appropriate
consideration for the public health and safety, and the
cultural, societal, and environmental considerations.

Conduct investigations of complex problems: Use

research-based knowledge and research methods including
design of experiments, analysis and interpretation of data,
and synthesis of the information to provide valid

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conclusions.

Modern tool usage: Create, select, and apply appropriate

techniques, resources, and modern engineering and IT tools including prediction and modeling to complex engineering activities with an understanding of the limitations.

The engineer and society: Apply reasoning informed by the

contextual knowledge to assess societal, health, safety, legal and cultural issues and the consequent responsibilities relevant to the professional engineering practice.

Environment and sustainability: Understand the impact of the

professional engineering solutions in societal and environmental contexts, and demonstrate the knowledge of, and need for sustainable development.

Ethics: Apply ethical principles and commit to professional

ethics and responsibilities and norms of the engineering practice.

Individual and team work: Function effectively as an

individual, and as a member or leader in diverse teams, and in multidisciplinary settings.

Communication: Communicate effectively on complex

engineering activities with the engineering community and with society at large, such as, being able to comprehend and write effective reports and design documentation, make effective presentations, and give and receive clear instructions.

Project management and finance: Demonstrate knowledge and

understanding of the engineering and management principles and apply these to one's own work, as a member and leader in a team, to manage projects and in multidisciplinary environments.

Life-long learning: Recognize the need for, and have the

preparation and ability to engage in independent and life-long learning in the broadest context of technological change.

Program Specific Outcomes (PSOs)

Analyze, Design and Implement application specific

Electronic systems for Analog and Digital domain, Communications, Signal and Image Processing Applications

Demonstrate the Computational and programming skills for

problem solving

Identify and Apply Domain specific tools for Design,

Analysis and Synthesis in the areas of VLSI and Embedded

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systems

M.Tech- VLSI System Design

Identify, formulate and analyze technical problems in areas

like semiconductor technologies, VLSI Signal verification
and Design verification and testing.

Design and implementation of VLSI architectures using FPGA

Use the techniques, skills, modern Electronic Design

Automation(EDA) tools to evaluate and analyze the
performance of the systems in VLSI domain.

Apply acquired knowledge from undergraduate engineering and

other disciplines to identify, formulate and present
solutions to technical problems related to various areas of
VLSI.

Learn advanced technologies and analyze complex problems in

the fields of VLSI.

Design and implementation of VLSI architectures using

FPGA/CPLD.

Addressing specific problems in the field of VLSI system

design in the form of mini projects, analysis and
interpretation of data and synthesis of information to
provide valid conclusions.

Use the techniques, skills, modern Electronic Design

Automation (EDA) tools, software and equipment necessary to
evaluate and analyze the systems in VLSI design
environments.

Understand and commit to professional ethics, social

responsibilities and norms of engineering practice.

Develop confidence for self-education and imbibe

professional values for lifelong learning.

Demonstrate effective oral and written communication skills

in accordance with technical standards.

Become knowledgeable about contemporary developments.

Ability to correct the mistakes effectively and learn from

them to become good leaders.

Understand the scenario of global business.

Design and implementation of VLSI architectures using FPGA.

M.Tech- Embedded Systems

Demonstrate outstanding analytical & technical skills to

evaluate , analyze and solve real time problems in Embedded
Systems.

Apply the acquired knowledge to solve engineering problems

to suit multidisciplinary situations.

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Undertake research and development projects in the field of

Embedded Systems.

Continue the personal development through professional study

and self learning.

Demonstrate their professional, ethical and social

responsibilities and contribute their part for addressing

various global issues.

Apply the acquired knowledge from undergraduate engineering

and other disciplines to identify, formulate and present

solutions to technical problems related to various areas of

Embedded Systems.

Learn advanced technologies and analyze complex problems in

the fields of Embedded System design along with the

fundamental concepts of engineering.

Addressing specific problems in the field of Embedded System

in the form of mini projects, analysis, and interpretation

of data and synthesis of information to provide valid

conclusions by considering societal and environmental

factors in the core areas of expertise.

Plan and conduct systematic study on a significant research

topic effective to the societal, health, legal and

environmental issues in the field of Embedded Systems.

Use the techniques, skills, modern Integrated Development

Environment (IDE) tools, Operating systems, software and

equipment necessary to evaluate and analyze the systems in

Real time environments.

Ability to manage team effectively and become good leaders.

Develop confidence and motivation for self education and

imbibe professional values for lifelong learning.

Become knowledgeable about contemporary developments by self

learning.

Identify, formulate and solve engineering problems in the

broad areas like System Design using Embedded Platforms,

Applications in Signal Processing, Machine Vision and

Communication Networks.

Use different software tools in the domain of Embedded

Systems, for developing applications based on ARM processor

core SoC and DSP processor.

Integrate multiple sub-systems to optimize performance of

Embedded Systems and excel in industry sectors related to

Embedded Systems domain.